

handle eight hardware threads at the same time. The POWER8 has a number of features which distinguishes it from its competitors. These include:

- First chip designed by IBM specifically for the big data and cloud computing market to handle vast amounts of data, processing it and storing it.
- Four times faster than the previous generation of POWER based chips, with four times the number of threads per core than the last generation
- 170% more bandwidth
- Linux-based ecosystem making it ideal for big data and cloud computing solutions
- CAPI (Coherent Attached Processor Interface) port support that enables the connection of GPUs, flash memory and FPGAs directly to systems
- Compatibility to scale from the current generation DDR3 memory controller to the DDR4 standard with the help of an external component acting as a memory buffer called 'Centaur'

IBM partnered with NVIDIA for their POWER8 chip to add hardware based Java acceleration on the GPU, speeding up JAVA code by up to eight times. IBM has also been releasing servers and computers based on the latest POWER8 chip since its announcement in 2014. 